

PROBABILITY AND RANDOM PROCESSES

Module 1

- Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, the Binomial probability distribution, the Poisson probability distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables.

Module 2

- Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables.

Module 3

- Limit theorems : Markov's Inequality, Chebyshev's Inequality, Strong Law of Large Numbers (Without proof), Central Limit Theorem (without proof), Stochastic Processes: Discrete-time process, Continuous-time process, Counting Processes, The Poisson Process, Interarrival times (Theorems without proof)

Module 4

- Markov Chains, Random Walk Model, Chapman–Kolmogorov Equations, Classification of States, Irreducible Markov chain, Recurrent state, Transient state, Long-Run Proportions. (Theorems without proof)

THEORY OF COMPUTATION

Module 1

- **Foundations (Linz, Hopcroft)**

Motivation for studying computability, need for mathematical modeling - automata, Introducing automata through simple models - On/Off switch, coffee vending machine. Three basic concepts: Alphabet, Strings, and Languages

- **Finite Automata (Linz, Hopcroft)**

Formal definition of a finite automaton, Deterministic Finite Automata (DFA), Regular languages, Nondeterminism (guess and verify paradigm), Formal definition of a nondeterministic finite automaton, NFA with epsilon transitions, Eliminating epsilon transitions (Proof not expected), Equivalence of NFAs and DFAs (Proof not expected) - The Subset Construction. DFA State Minimization, Applications of finite automata - text search, keyword recognition

Module 2

- **Regular Expressions (Linz)**

The formal definition of a regular expression, Building Regular Expressions, Equivalence with finite automata (Proof not expected) - Converting FA to Regular Expressions, Converting Regular Expressions to FA, Pattern Matching and Regular Expressions, Regular grammar, Equivalence with FA - Conversion in both directions

- **Properties of Regular Languages (Linz)**

Closure and Decision Properties of Regular Languages (with proofs), The Pumping Lemma for Regular Languages (with formal proof), Pumping lemma as a tool to prove non regularity of languages

- **Context-Free Grammars and Applications (Linz)**

Formal definition of a context-free grammar, Designing context-free grammars, Left-most and Rightmost Derivations Using a Grammar, Parse Trees, Ambiguous Grammars, Resolving ambiguity, Inherent ambiguity, CFGs, and programming languages

Module 3

- **Pushdown Automata (Linz)**

Formal definition of a pushdown automaton, DPDA and NPDA, Examples of pushdown automata

Equivalence NPDAs and CFGs (Proof not expected) - conversions in both directions

- **Simplification of Context-Free Languages (Linz)**

Elimination of useless symbols and productions, Eliminating epsilon productions, Eliminating unit productions, Chomsky normal form, Greibach normal form,

- **Properties of Context-Free Languages (Linz)**

The Pumping Lemma for Context-Free Languages (with formal proof), Closure and Decision Properties of Context-Free Languages (with formal proofs)

Module 4

- **Turing Machines (Kozen)**

The formal definition of a Turing machine, Examples of Turing machines - Turing machines as language acceptors, Turing machines as computers of functions, Variants of Turing Machines (Proofs for equivalence with basic model not expected), Recursive and recursively enumerable languages

- Chomskian hierarchy, Linear bounded automaton as a restricted TM.

- **Computability (Kozen)**

Church Turing thesis, Encoding of TMs, Universal Machine and Diagonalization, Reductions, Decidable and Undecidable Problems, Halting problem, Post Correspondence Problem and the proofs for their undecidability.

DATA STRUCTURES AND ALGORITHMS

Module 1

- **Basic Concepts of Data Structures**

Definitions; Data Abstraction; Performance Analysis - Time & Space Complexity, Asymptotic Notations; Polynomial representation using Arrays, Sparse matrix (Tuple representation); Stacks and Queues - Stacks, Multi-Stacks, Queues, Circular Queues, Double Ended Queues; Evaluation of Expressions- Infix to Postfix, Evaluating Postfix Expressions.

Module 2

- **Linked List and Memory Management**

Singly Linked List - Operations on Linked List, Stacks and Queues using Linked List, Polynomial representation using Linked List; Doubly Linked List; Circular Linked List; Memory allocation - First-fit, Best-fit, and Worst-fit allocation schemes; Garbage collection and compaction.

Module 3

- **Trees and Graphs**

Trees: - Representation Of Trees; Binary Trees - Types and Properties, Binary Tree Representation, Tree Operations, Tree Traversals; Expression Trees; Binary Search Trees - Binary Search Tree Operations; Binary Heaps - Binary Heap Operations, Priority Queue.

Graphs: - Definitions; Representation of Graphs; Depth First Search and Breadth First Search; Applications of Graphs - Single Source All Destination.

Module 4

- **Sorting and Searching**

Sorting Techniques: - Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort.

Searching Techniques: - Linear Search, Binary Search, Hashing - Hashing functions : Mid square, Division, Folding, Digit Analysis; Collision Resolution : Linear probing, Quadratic Probing, Double hashing, Open hashing.

OBJECT ORIENTED PROGRAMMING

Module 1

- **Introduction to Java:**

Structure of a simple java program; Java programming Environment and Runtime Environment (Command Line & IDE); Java compiler; Java Virtual Machine; Primitive Data types and Wrapper Types; Casting and Autoboxing; Arrays; Strings; Vector class; Operators - Arithmetic, Bitwise, Relational, Boolean Logical, Assignment, Conditional (Ternary); Operator Precedence; Control Statements - Selection Statements, Iteration Statements and Jump Statements; Functions; Command Line Arguments; Variable Length Arguments; Classes; Abstract Classes; Interfaces. [Use proper naming conventions]

- **OOP Concepts :-**

Data abstraction, encapsulation, inheritance, polymorphism, Procedural and object oriented programming paradigm; Microservices.

- **Object Oriented Programming in Java :-**

Declaring Objects; Object Reference; Introduction to Methods; Constructors; Access Modifiers; this keyword.

Module 2

- **Polymorphism :-**

Method Overloading, Using Objects as Parameters, Returning Objects, Recursion.

- Static Members, Final Variables, Inner Classes.

- **Inheritance** - Super Class, Sub Class, Types of Inheritance, The super keyword, protected Members, Calling Order of Constructors.

- Method Overriding, Dynamic Method Dispatch, Using final with Inheritance.

Module 3

- **Packages and Interfaces –**

Packages - Defining a Package, CLASSPATH, Access Protection, Importing Packages.

Interfaces - Interfaces v/s Abstract classes, defining an interface, implementing interfaces, accessing implementations through interface references, extending interface(s).

- **Exception Handling** - Checked Exceptions, Unchecked Exceptions, try Block and catch Clause, Multiple catch Clauses, Nested try Statements, throw, throws and finally, Java Built-in Exceptions, Custom Exceptions.

- Introduction to design patterns in Java : Singleton and Adaptor.

Module 4

- SOLID Principles in Java (<https://www.javatpoint.com/solid-principles-java>)
- **Swings fundamentals** – Overview of AWT, Swing v/s AWT, Swing Key Features, Model View Controller (MVC), Swing Controls, Components and Containers, Swing Packages, Event Handling in Swings, Swing Layout Managers, Exploring Swings–JFrame, JLabel, The Swing Buttons, JTextField.
- **Event handling** – Event Handling Mechanisms, Delegation Event Model, Event Classes, Sources of Events, Event Listener Interfaces, Using the Delegation Event Model.
- **Developing Database Applications using JDBC** – JDBC overview, Types, Steps, Common JDBC Components, Connection Establishment, SQL Fundamentals [For projects only] - Creating and Executing basic SQL Queries, Working with Result Set, Performing CRUD Operations with JDBC.

DIGITAL ELECTRONICS AND LOGIC DESIGN

Module 1

- **Introduction to digital Systems** :- Digital abstraction
- **Number Systems** – Binary, Hexadecimal, grouping bits, Base conversion; Binary Arithmetic – Addition and subtraction, Unsigned and Signed numbers; Fixed-Point Number Systems; Floating-Point Number Systems
- **Basic gates**- Operation of a Logic circuit; Buffer; Gates - Inverter, AND gate, OR gate, NOR gate, NAND gate, XOR gate, XNOR gate; Digital circuit operation - logic levels, output dc specifications, input dc specifications, noise margins, power supplies; Driving loads - driving other gates, resistive loads and LEDs.
- **Verilog (Part 1)** :-
HDL Abstraction; Modern digital design flow - Verilog constructs: data types, the module, Verilog operators.

Module 2

- **Combinational Logic Design:** –
Boolean Algebra - Operations, Axioms, Theorems; Combinational logic analysis - Canonical SOP and POS, Minterm and Maxterm equivalence; Logic minimization - Algebraic minimization, K-map minimization, Dont cares, Code convertors.
- **Modeling concurrent functionality in Verilog:-**
Continuous assignment - Continuous Assignment with logical operators, Continuous assignment with conditional operators, Continuous assignment with delay.

Module 3

- **MSI Logic and Digital Building Blocks**
MSI logic - Decoders (One-Hot decoder, 7 segment display decoder), Encoders, Multiplexers, Demultiplexers; Digital Building Blocks - Arithmetic Circuits - Half adder, Full adder, half subtractor, full subtractor; Comparators.
- **Structural design and hierarchy** - lower level module instantiation, gate level primitives, user defined primitives, adding delay to primitives.

Module 4

- **Sequential Logic Design** :- Latches and Flip-Flops- SR latch, SR latch with enable, JK flipflop, D flipflop, Register Enabled Flip-Flop, Resettable Flip-Flop. Sequential logic timing considerations; Common circuits based on sequential storage devices - toggle flop clock divider, asynchronous ripple counter, shift register.
- **Finite State Machines** :-
Finite State Machines - logic synthesis for an FSM, FSM design process and design examples; Synchronous Sequential Circuits - Counters;

- **Verilog (Part 2) :** -
Procedural assignment; Conditional Programming constructs; Test benches; Modeling a D flipflop in Verilog; Modeling an FSM in Verilog.